

Impact of micro-credential learning on students' perceived employability: the mediating role of human capital

Micro-
credential
learning

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Abstract

Purpose – This paper examines the influence of micro-credential learning on students' perceived employability. In addition, the study aims to explore different components that will help students to gain knowledge, enhance their careers and develop their human capital (social, cultural and scholastic capital). Hence, the study also analyzed the mediating role of human capital on the aforementioned association.

Design/methodology/approach – Explanatory research was conducted by utilizing a correlational research design. A questionnaire comprising of closed-ended items was utilized in the study. The data was analyzed by employing PLS-SEM technique.

Findings – Our findings stipulate that micro-credential learning is an essential component to improve students' perceived employability. The study identified that micro-credential programs have a positive relationship with students' perceived employability. Moreover, the findings that micro-credential learning significantly impacts students' human capital, i.e. cultural, social and scholastic capital. Additionally, human capital acts as a significant mediator in the relationship between micro-credential learning and students' perceived employability. Moreover, participation in micro-credential learning can ensure that students can identify diverse career directions, seek information about the labor market and educational system, attain relevant qualifications for their vocations, and develop a plan for their future.

Originality/value – Micro-credential programs are short and focused educational programs that offer specialized knowledge and skills in a particular area. These programs are becoming increasingly popular in the modern workforce to upskill or reskill quickly and efficiently. However, lack of empirical evidence is the ultimate gap in determining the importance of micro-credential learning; as the limited literature is unable to determine the importance of MCL on students' perceived employability. Thus, the study identifies the impact of micro-credential learning on students' perceived employability.

Keywords Micro-credential learning, Social capital, Scholastic capital, Cultural capital,

Perceived employability

Paper type Research paper



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1. Introduction

Recently, online learning has extensively changed students' learning processes. The first online learning system was introduced in the 1980s which was referred to as e-Learning (Keegan, 1986); it allows students to acquire knowledge at their own pace with proper resources without any geographical barrier. The majority of people were not familiar with the e-learning system; nevertheless, it gained considerable popularity in society after the pandemic. Accordingly, higher education adopts the advanced learning pattern and changes the traditional learning approach. Furthermore, the pandemic has changed traditional learning approaches and significantly augmented the concept of online learning, short courses, certificated courses and micro-credential programs (Cascio, 2019).

After the pandemic, short courses, certificated courses and micro-credential programs have gained immense importance in this society. Micro-credentials are mini-certifications that demonstrate the developed skill or experience of the students in a certain subject area; it also assists students to acquire more subject knowledge (Pickard, 2018). These courses help students to improve their human capital. Moreover, these courses also provide extensive exposure to the employees to get connected with other professionals and mentors to acquire information regarding expectations or different areas in the labor market.

Micro-credential learning substantially improves the planning of instruction and assists students in obtaining the qualifications, skills or experience required for their future employability. According to Rothwell and Arnold (2007), employability is the capacity of an individual to obtain the desired job or maintain their current employment. Higher Education (HE) has a substantial role to prepare students for finding suitable jobs; as they are aware of the growing market competition in the jobs market. Byrne (2022) established that the job market is impacted by tumultuous circumstances such as the COVID-19 pandemic. As a result, numerous stakeholders have given considerable attention to employability. Therefore, the identification of an appropriate fit between higher education and businesses is mandatory to increase students' employability in order to connect graduates' skills and the need of employers (Bennett *et al.*, 2016). Micro-credentials are an essential component of the digital credentialing ecosystem achievable by networks of interest created by digital communications technology; It allows people to share information about learner comprehension and their capability in the market. Micro-credentials program helps students to improve potential skills and use them to satisfy the demands of the modern workforce (Milligan and Kennedy, 2017).

Correspondingly, micro-credentials enable students to document a range of experiences and abilities acquired during the program and present it as extra certifications along with the conventional transcripts. To encourage the stakeholders to trust the skills obtained by micro-credentials, a digital credential system makes pertinent information about the micro-credential program (contents, quality, outcomes, assessment, workload) (Schilirò, 2021); usually, accessible and appropriate to use. According to Schwab and Zahidi (2020), The World Economic Forum predicted a shift in how humans and machines perform work might lead to the loss of 85 million jobs by 2025; it highlighted the potential difficulties that workers and recent graduates may encounter due to the effect of the pandemic on economies.

Moreover, few professions are transforming rapidly with the changes in the workplace. Promptly, the success of recent graduates will be determined by their technical and soft skills; hence, micro-credential courses reduce the employment gap between higher education and the workforce (Ruddy and Ponte, 2019). These courses provide technical training to assist students in overcoming market hurdles by acquiring in-demand job skills (Crow, 2017).

Additionally, micro-credential programs also invest in the learner's human capital (social capital, cultural capital and scholastic capital) which directly helps employability of the students (Donald *et al.*, 2017). Scholastic capital is an educational achievement; it has a favorable effect on a person's career achievements. Scholastic capital confers and analyzes individual information and suggests the significance of work-integrated learning that will assist students in future

employment. Thus, these programs help students in multiple ways; to understand the labor market, develop a sense of identity by participating in career counseling events, get a chance to connect with graduates and employers, and take career skills courses that will help them in the future (Bridgstock *et al.*, 2019). Furthermore, other earlier studies (Donald *et al.*, 2019; Baruch *et al.*, 2005) affirm that this capital is a significant predictor in the learning process. This accentuates that students' employability confidence increases as their academic competencies increase which eventually helps them to achieve career objectives. In addition, establishing career goals can assist students in effectively building their human capital by acquiring relevant practical knowledge and expertise (Direnzo *et al.*, 2015). As a result, the candidates acquiring impressive scholastic capital will be preferred in job.

Moreover, the micro-credential program also cultivates students' social capital by incorporating community service activities (Donald *et al.*, 2017); as it also assists students in building networks that can benefit them in their professional endeavors. In addition, social connections are a crucial component that can influence employers' hiring decisions by favorably recommending candidates; these connections also provide resources beyond the job applicant's knowledge and skills (Stiff and Vugt, 2008). The micro-credential program seeks to increase students' employability by developing relevant knowledge and skills required in the workplace. Students can also develop networking skills via project-based learning, academic partnerships between universities and corporate sectors, alumni relationships, etc. Moreover, this program also enhances students' cultural capital by developing their social values (Steinfeld *et al.*, 2008; Donald *et al.*, 2017). DiMaggio (1982) validated that teachers pay considerable attention to promote and reward students with high cultural capital or high involvement in elite societies. An individual with a high cultural capital may attract the attention of employers in the HE sectors. Therefore, micro-credential programs also encourage students to develop their cultural capital as it substantially boosts students' confidence in their future jobs (Watts, 2006).

Accordingly, micro-credential programs offer individuals the opportunity to quickly gain specific skills and knowledge in a particular area. These programs are shorter and more focused than traditional degree programs; thus, allowing individuals to upskill or reskill efficiently (Robinson *et al.*, 2020). However, the effectiveness of micro-credential programs in improving students' perceived employability is not properly and extensively examined.

Several studies have investigated different aspects regarding the impact of micro-credential programs on employability. For instance, González Moreno *et al.* (2019) determined that micro-credentials can help individuals demonstrate their knowledge and skills to potential employers. Another study by Bosch-Sijtsema *et al.* (2011) ascertained that micro-credentials can enhance individuals' perceived employability and increase their chances of getting hired. Despite existing research on the impact of micro-credential programs on employability, further investigation is essential to attain comprehensive insight of their effects on students' perceived employability. Maina *et al.* (2022) investigated impact of micro-credentials on "students' perceived employability" in East African universities; findings validate micro-credentials can increase students' confidence in their job prospects and improve their perceived employability.

According to the authors' best knowledge, no study has analyzed the impact of micro-credential learning on perceived employability. The prior literature has mainly highlighted the effect of career development education on students' perceptions regarding their employability (Ho *et al.*, 2022). Similarly, Ma and Bennett (2021) investigate link between perceived employability, academic engagement and stress of Chinese higher education. Similarly, Maina *et al.* (2022) reveal an appropriate micro-credentialing approach that will help in recognizing and developing HE employable skills. Similarly, Ahmed and Jassim (2021) examine students' and teachers' attitudes toward adopting micro-credentials in higher education to improve the standard of living and employability skills. However, no study has yet highlighted the concept of micro-credential learning on students' perceived employability in the same conceptual framework. Therefore, this research will fill the existing gap in the present literary work by

highlighting the importance of short courses, online learning courses and micro-credential courses (as the pandemic has entirely transformed the whole learning system).

Moreover, our research aims to address the following issues. Firstly, the lack of understanding about the influence of micro-credential programs on students' perceived employability, and their ability to secure jobs in the future. Secondly, Students' perceived employability is an important factor in job search and career development; nevertheless, it is unclear how micro-credential programs impact this perception. Therefore, the study aims to provide insights into the effectiveness of these programs in improving students' job prospects and identify areas for improvement in micro-credential program design and implementation. It will also contribute to the existing awareness on micro-credential programs and its impact on students' perceived employability, which will help in making effective policy decisions and adopting educational practices.

The study is systematized as follows: [section 1](#) comprises introduction of the study. [Section 2](#) confers relevant literature review. Additionally, third section encloses methodology of study while fourth section contains the discussion and results of the study. Finally, conclusion and implications are included in the fifth section.

2. Literature review

2.1 Theoretical background

The current study underpinned "Human Capital Theory-(HCT)" to explain relationship between Micro-Credentials, Human Capital (HC) and Students' Perceived Employability (PE). "Human Capital Theory" is an economic theory developed by Gary [Becker \(1964\)](#). It considers education and training as an investment in individuals' human capital. According to this theory, individuals can improve their productivity and earning potential through education and training, which enhances their skills, knowledge and abilities ([Brown et al., 2020](#)). HCT suggests that education and training are effective methods for individuals to enhance their skills and knowledge, also such investments in human capital can increase their productivity and employability ([Becker, 1964](#)). In addition, micro-credentials can develop specific skills and knowledge among learners that are relevant to specific job descriptions or industries ([Walters, 2004](#)). Therefore, micro-credentials can significantly enhance students' HC which can enhance their employability.

Moreover, HCT Theory highlights importance of education and training as a means of enhancing individuals' skills, productivity and earning potential, as well as promoting economic growth and development. These programs are more reasonable and augment quick development of human capital as compared to traditional education ([Phelan and Glackin, 2020](#)). Accordingly, students with high human capital are likely to have higher competitive advantage in labor market and have high perceived employability. Correspondingly, Micro-credentials help individuals to acquire specific skills and knowledge in a short period, which substantially enhance their human capital and increase their perceived employability. Therefore, micro-credentials can provide evidence of an individual's commitment to continuous learning and skill development, which can be attractive to employers who value lifelong learning and adaptability. Therefore, by investing in their human capital through micro-credential learning, students can enhance their perceived employability and increase their chances of securing employment in their desired field. In accordance with the HCT, the study is analyzing the impact of micro-credentials on human capital and students' perceived employability.

2.2 Micro-credential learning and student perceived employability

A micro-credential is an accredited documentation of the achievements and acquired knowledge offered by institutions or companies. These certifications enhance academic profile in addition with experiences and qualities including interpersonal skills, extracurricular and voluntary

activities that are not represented on academic transcripts or resumes (Braxton-Stewart, 2019). Capone *et al.* (2021) examine the relationships among students' (PE), self-efficacy, ambition, organizational dedication and career planning, along with students' engagement, mental health and academic weariness in the pandemic period. Findings highlight students experience high uncertainty regarding careers and employability. Despite the effects of COVID-19, students exhibit high participation, professional ambitions and good self-efficacy. Accordingly, another empirical study by Wei and Chou (2020) provides a descriptive structural model to ascertain the influence of students' attitudes and readiness toward learning performance and course satisfaction in Taiwan. The results revealed a positive impact on Student's performance in online discussions and course satisfaction. Moreover, findings determine that students' self-reassurance to use computers and Internet for online learning had a significant impact on their perceptions of online learning readiness, online discussion scores and satisfaction with courses. Additionally, the study discovered that Internet and computer self-efficacy had a mediating effect on all the above-mentioned variables, i.e. students' attitudes, readiness, learning performance and course satisfaction.

Similarly, Ho *et al.* (2022) investigate the relationship of students' perceived employability (SPE) and career development learning (CDL) with the mediation of Human Capital (HC). Finally, Maina *et al.* (2022) accentuated the significance of graduates' employment abilities and a transparent procedure for recognition. The Employability Skills Micro-credentialing (ESMC) methodology was employed in this study. The results recommend that employability skills must be incorporated into the academic curriculum and educational practices. Hence, the subsequent hypothesis is developed:

- H1. Micro-credential learning has a positive and significant impact on student perceived employability.

2.3 Micro-credential learning and human capital

Human capital is a significant asset that include personal attributes or economic value of an individual which are; for instance, knowledge, abilities, job experiences and education (Østergaard and Marinova, 2018). According to Clarke (2018), human capital is subdivided into three capital, i.e. scholastic, cultural and social capital. Moreover, Watts and Sultana (2004) synthesize three coordinated evaluations of "National Career Policies" by the OECD, the World Bank and European commission. The findings highlight ten characteristics of the lifelong system along with the seven fundamental issues that required stakeholder attention. The findings revealed that career counseling through micro-credential learning helps students to attain sustainable employability in the labor market. A study by Gilani (2020) analyzed the association of students' social capital and graduates' employment in the UK. The findings revealed that these programs augment students' professional networks and make social contacts which will help in the future.

Additionally, Batistic and Tymon (2017) conceptually and empirically investigated networking behavior and job readiness after higher education. The results highlight that networking increases access to information and resources, which directly/significantly increases perceptions of one's internal and external employability. Finally, Wheelahan and Moodie (2021) ascertain that micro-credential learning should incorporate the application of HCT with twenty-first-century skills in the HE curricula to enhance its compatibility with the demands of labor markets. Results established micro-credentials as a powerful tool for "disciplining" higher education curriculum to align it with the requirements of the workplace that increase the perspective of human capital; thus, people are willing to invest in themselves by realizing demands of future markets. Thus, the following hypothesis has been formulated:

- H2. Micro-credential learning has a positive and significant impact on cultural capital.

H3. Micro-credential learning has a positive and significant impact on social capital.

H4. Micro-credential learning has a positive and significant impact on scholastic capital.

2.4 Human capital and students' perceived employability

Perceived employability refers to a set of accomplishments, knowledge, abilities and personal characteristics that increase graduates' possibilities of job in the desired professions (Clarke, 2017; Costea *et al.*, 2012). Moreover, Donald *et al.* (2017) examined the influence of human capital on undergraduates' understanding of their employability (social capital, cultural capital, scholastic capital and skills) on undergraduates' and views their employability. The findings revealed that the utilization of well-informed tactics for hiring, preparing, attracting and retaining graduates can benefit all parties involved by improving undergraduates' employability perceptions. Trimble and Kmec (2011) explore the importance of social networks in accessing job information and finding employment in the United States. The result revealed that relying on social networks to obtain work might be futile for the workplace; as social networks are not effective for every job seeker. Accordingly, Finch *et al.* (2013) evaluate elements that affect university graduates' employment. The findings represent that firms prioritize soft skills of recent graduates during the recruiting process and give least consideration to the academic reputation.

Similarly, Caballero *et al.* (2021) examine a comprehensive model of perceived employability, HC, i.e. cultural and social capital among university graduates. Outcome indicates labor market, university networks, self-assurance in searching for employment opportunities and higher education are crucial elements in graduates' employability judgments. Finally, Donald *et al.* (2017) analyze college students' perception of their employability; It investigates the effects of human capital, i.e. talents, market-value, social, cultural, psychological and academic capital. The outcome demonstrates improvements in contemporary career theory and human capital theory to proceed from higher education to the workforce. Increased knowledge about undergraduates' perceptions of their employability will enhance the ability to better prepare, recruit, hire and retain graduates which will ultimately benefit all stakeholders. Ma and Bennett (2021) highlighted the connection with students' stress, academic engagement and employability perceptions. The findings imply that self-perceptions of employability are adversely correlated with perceived stress while positively correlated with students' academic involvement. The pathways were mediated by perceived employability. The following hypotheses are proposed:

H5. Cultural capital has a positive and significant impact on students' perceived employability.

H6. Social capital has a positive and significant impact on students' perceived employability.

H7. Scholastic capital has a positive and significant impact on students' perceived employability.

2.5 Mediating variables

The above-mentioned literature has established that "Human Capital (HC)" influence students' perceived employability and micro-credential learning. While different studies have acknowledged that cultural, social and scholastic capital significantly enhance students' confidence in their perceived employability. Similarly, a study has examined psychological capital and career outcomes along with the mediation of perceived employability and career engagement (Baluku *et al.*, 2021). In addition, multiple studies have analyzed the interceding role of perceived employability (Ho *et al.*, 2022), employability model and employability

(Baluku *et al.*, 2021; Ma and Chen, 2022). Therefore, this study will also evaluate the mediating role of human capital (i.e. cultural, social and scholastic capital) between students' (PE) and micro-credential learning. Hence, we proposed the following:

- H8. Cultural capital mediates the relationship between micro-credential learning and students' perceived employability.
- H9. Social capital mediates the relationship between micro-credential learning and students' perceived employability.
- H10. Scholastic capital mediates the relationship between micro-credential learning and students' perceived employability.

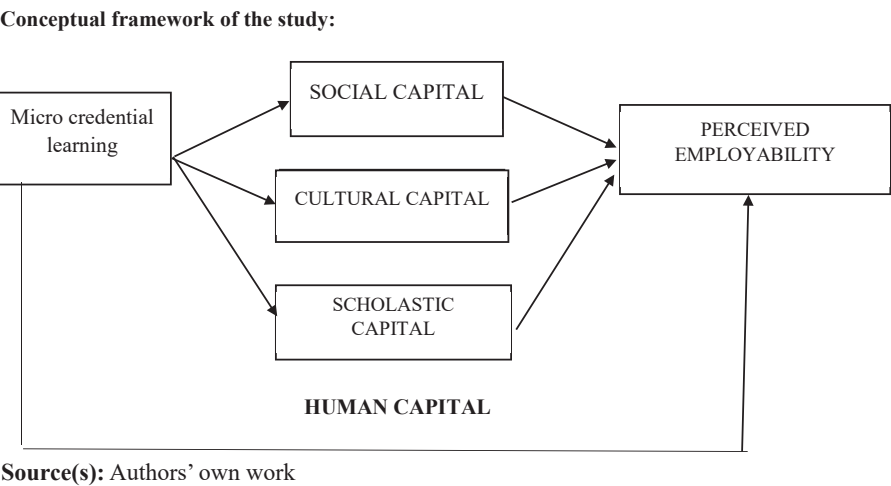
3. Methodology and data

The recent study utilized a quantitative research approach by using the "Explanatory Research" method to further investigate the dimension of the study. This study has a correlational research design, as it is used to investigate the relationship between the variables. Moreover, a correlational study is used to determine whether there is a negative, positive, or no relation among the variables. The study measures the influence of micro-credential learning on students' perceived employability and human capital, i.e. social capital, cultural capital and scholastic capital. Moreover, the study also examines the mediating role of human capital (social capital, cultural capital, scholastic capital) between the micro-credential learning students' perceived employability. The study utilized a self-administered questionnaire with closed-ended items. The data was collected from 234 students of higher education institutes in Karachi (Pakistan) by using a convenience sampling technique. As this technique helps in collecting data from a random pool of students. Furthermore, every measurement used in the questionnaire is taken from earlier and pertinent studies and graded on a five-point Likert scale (ranging from strongly disagree 1 to strongly agree 5).

Micro-credential learning was assessed by a close-ended questionnaire containing 26 items. These items examine Micro-Credential Learning, Scholastic Capital, Cultural Capital, Social Capital and Student Perceived Employability. The framework developed by Wei and Chou (2020) comprises six items of micro-credential learning. In addition, Scholastic Capital and cultural capital were examined by three items developed by Baruch *et al.* (2005) and two items by Jackson and Chapman (2012). Similarly, social capital was evaluated by two items proposed by Baruch *et al.* (2005) and three items by Jackson and Chapman (2012). While Student Perceived Employability was assessed by the six items developed by Ma and Bennett (2021). All these items were adapted from Wei and Chou (2020), Baruch *et al.* (2005), and Jackson and Chapman (2012), respectively.

The study has used the PLS-SEM technique for data analysis by using SmartPLS 3 to test the hypotheses in the current study. PLS-SEM is a useful tool to examine, describe and evaluate the constructed hypothesis. It enables the ongoing assessment of numerous dependent and independent factors. Data analysis was divided in two stages to complete PLS-SEM. Initially, the study evaluated the measurement model followed by the evaluation of the structural model. The study examined construct and discriminant validity criteria for the measurement model along with the importance of the path coefficients for the structural model. According to Ringle *et al.* (2015), Structural Equation Modeling (SEM) is a good statistical technique to examine the validity of the hypothesis. The variables are examined by using variance-based methodology. It enables the ongoing assessment of numerous dependent and independent factors, and facilitates the estimation of the direct and indirect consequences of the structure by using PLS-SEM utilizing Smart PLS software (Chin, 1998; Henseler *et al.*, 2009) (see Figure 1).

Figure 1.
The conceptual
research framework



4. Empirical analysis

4.1 Descriptive statistics

The sample consists of 234 students of higher education. [Table 1](#) illustrates descriptive statistics of research; the demographic represent 51.7% of responders were male and 48.3% were female. Moreover, the frequency for males is 121 and 113 for females, respectively. In terms of age, 63.2% of participants were between the ages of 18–25, 30.3% percent were between the ages of 26–33, 5.6% of the population was between the ages of 34–41 while 9% of the participants were aged between 42 and above. Furthermore, the frequency of the age bracket from 18 to 25 is 148, from 26 to 33 is 71, from 34 to 41 is 13, and from 42 onwards is 2, respectively.

Items	Frequency	Percent
<i>Age</i>		
18–25	148	63.2
26–33	71	30.3
34–41	13	5.6
42 and above	2	0.9
Total	234	100.0
<i>Gender</i>		
Male	121	51.7
Female	113	48.3
Total	234	100.0
<i>Education</i>		
Undergraduate	142	60.7
Graduate	51	21.8
Postgraduate	37	15.8
Others	4	1.7
Total	234	100.0

Table 1.
Demographic details **Source(s):** Authors' own work

4.2 Measurement model

4.2.1 Convergent validity. In this section, convergent validity, discriminant validity, construct reliability, and item reliability have been examined. Table 2 displays the outcomes of loading, Cronbach's alpha, composite reliability and average variance extract (AVE). The findings indicate that all the values for Cronbach's alpha and composite dependability are greater than 0.7; hence, meeting the standard (Straub, 1989). Composite reliability is the next metric that establishes all values are higher than 0.7 and fulfill the requirements of Churchill (1979). Our research met the requirements of Fornell and Larcker's (1981) standards; as all values in the final test AVE were found to be above 0.5. Moreover, we assessed the convergent validity of our research by utilizing AVE. Finally, we evaluated the convergent validity of our research using AVE.

4.2.2 Discriminant validity. Table 3 represents the results of the discriminant validity evaluation; subsequent to the cross-loading analysis AVE was extracted to quantify discriminant validity. Accordingly, the average variance must be higher than the correlation between the variables, i.e. all diagonal values are higher than non-diagonal values which is the criteria of Fornell and Larcker (1981). Thus, the table indicates the square root of AVE is meeting the requirements/criteria; as displayed diagonally.

Table 4 illustrates the cross-loadings of all variables, the findings highlight that each item is higher in their respective constructs than the related constructs. Cross-loading differences also exceed the recommended 0.1 level (Gefen and Straub, 2005).

Table 5 represents the heterotrait-monotrait correlation ratio must be smaller than 0.85 for all values; thus, satisfying the predetermined criterion (Henseler et al., 2015).

	Items	Loadings	Cronbach's alpha	Composite reliability	Average variance extracted
CC	CC1	0.737	0.818	0.873	0.580
	CC2	0.795			
	CC3	0.818			
	CC4	0.717			
	CC5	0.735			
MCL	MCL1	0.730	0.779	0.858	0.602
	MCL2	0.816			
	MCL3	0.774			
	MCL4	0.780			
PE	PE1	0.828	0.825	0.877	0.588
	PE2	0.784			
	PE3	0.740			
	PE4	0.756			
	PE5	0.721			
SCHC	SCHC1	0.860	0.918	0.939	0.754
	SCHC2	0.840			
	SCHC3	0.891			
	SCHC4	0.883			
	SCHC5	0.865			
SOC	SOC1	0.754	0.772	0.853	0.593
	SOC2	0.788			
	SOC3	0.743			
	SOC4	0.793			

Note(s): MCL = Micro-credential learning, PE = Perceived employability, CC = Cultural Capital, SOC = Social Capital, SCHC = Scholastic Capital

Source(s): Authors' own work

Table 2.
Measurement model
results

Table 3.
Fornell-Larcker
criterion

4.3 Structural model

The structural model is utilized to examine the relationships between the variables in our research model. Accordingly, regression analysis is used to assess the structural model. We also considered 0.1 or 10%, as the standard significance level for the *p*-values; as the *p*-value should not be more significant than 0.01 (1%), 0.05 (5%) or 0.1 (10%) for any hypothesis to be accepted. As demonstrated in Table 6, all determined hypotheses were accepted. Additionally, the findings of mediating variables are demonstrated in Table 7.

	CC	MCL	PE	SCHC	SOC
CC	0.761				
MCL	0.507	0.776			
PE	0.577	0.574	0.767		
SCHC	0.419	0.401	0.483	0.868	
SOC	0.664	0.572	0.655	0.404	0.770

Note(s): MCL = Micro-credential learning, PE = Perceived employability, CC = Cultural Capital, SOC = Social Capital, SCHC = Scholastic Capital
Source(s): Authors’ own work

Table 4.
Loadings and cross
loadings

	CC	MCL	PE	SCHC	SOC
CC1	0.737	0.360	0.385	0.338	0.486
CC2	0.795	0.450	0.450	0.322	0.542
CC3	0.818	0.395	0.458	0.338	0.517
CC4	0.717	0.295	0.464	0.274	0.476
CC5	0.735	0.417	0.437	0.321	0.501
MCL1	0.356	0.730	0.413	0.210	0.452
MCL2	0.437	0.816	0.510	0.354	0.465
MCL3	0.359	0.774	0.388	0.306	0.466
MCL4	0.414	0.780	0.461	0.364	0.396
PE1	0.516	0.563	0.828	0.459	0.595
PE2	0.440	0.449	0.784	0.374	0.486
PE3	0.404	0.400	0.740	0.265	0.476
PE4	0.355	0.365	0.756	0.369	0.462
PE5	0.478	0.393	0.721	0.359	0.475
SCHC1	0.369	0.339	0.420	0.860	0.379
SCHC2	0.329	0.352	0.449	0.840	0.343
SCHC3	0.432	0.405	0.416	0.891	0.397
SCHC4	0.369	0.343	0.428	0.883	0.308
SCHC5	0.307	0.293	0.374	0.865	0.318
SOC1	0.527	0.349	0.486	0.326	0.754
SOC2	0.477	0.525	0.546	0.279	0.788
SOC3	0.493	0.409	0.477	0.338	0.743
SOC4	0.554	0.458	0.505	0.310	0.793

Note(s): MCL = Micro-credential learning, PE = Perceived employability, CC = Cultural Capital, SOC = Social Capital, SCHC = Scholastic Capital
Source(s): Authors’ own work

4.4 Discussion

Tables 6 and 7 represent the results of first-order analyses that comprise results of Path Analysis and Mediation Analysis. Ten hypotheses have been developed in the study. In Table 6, the findings indicate that seven hypotheses are accepted: H1, H2, H3, H4, H5, H6 and H7. In addition, Table 7 represents the findings of mediating variables including H8, H9 and H10.

The first hypothesis is accepted (MCL → PE) that indicates a positive relation between micro-credential learning and perceived employability ($\beta = 0.221, p < 0.004$). Similarly, the results of Donald *et al.* (2021) conclude micro-credential learning significantly impacts perceived employability. The results validate the findings of Fényes *et al.* (2021) and Geng *et al.* (2022), the study indicates that students volunteer and participate in activities and programs for professional development and career opportunities. In addition, Wheelahan and Moodie (2021) established that micro-credential has the potential to align higher education with the putative market and its requirements. Therefore, micro-credential learning significantly improves students' skills, which motivates them to be in a good position in any organization.

The second hypothesis (MCL → CC) reveals a significant relation between micro-credential learning and cultural capital ($\beta = 0.514, p < 0.000$). Bourdieu and Richardson (1986) affirmed a positive relationship between micro-credential learning and cultural capital. The

	CC	MCL	PE	SCHC	SOC
CC					
MCL	0.629				
PE	0.696	0.702			
SCHC	0.481	0.467	0.546		
SOC	0.837	0.731	0.813	0.481	

Note(s): MCL = Micro-credential learning, PE = Perceived employability, CC = Cultural Capital, SOC = Social Capital, SCHC = Scholastic Capital

Source(s): Authors' own work

Table 5.
Heterotrait–Monotrait
ratio (HTMT)

Hypothesis	Regression path	Effect type	β -Coeff	p values	Remarks
H1	MCL → PE	Direct Effect	0.221	0.004	Supported
H2	MCL → CC	Direct Effect	0.514	0.000	Supported
H3	MCL → SOC	Direct Effect	0.576	0.000	Supported
H4	MCL → SCHC	Direct Effect	0.402	0.000	Supported
H5	CC → PE	Direct Effect	0.163	0.043	Supported
H6	SOC → PE	Direct Effect	0.348	0.000	Supported
H7	SCHC → PE	Direct Effect	0.184	0.000	Supported

Table 6.
Results of path
analysis

Hypothesis	Regression path	Effect type	β -Coeff	p values	Remarks
H8	MCL → CC → PE	Indirect Effect	0.084	0.073	Partial Mediation
H9	MCL → SOC → PE	Indirect Effect	0.200	0.000	Partial Mediation
H10	MCL → SCHC → PE	Indirect Effect	0.074	0.002	Partial Mediation

Note(s): MCL = Micro-credential learning, PE = Perceived employability, CC = Cultural Capital, SOC = Social Capital, SCHC = Scholastic Capital

Source(s): Authors' own work

Table 7.
Mediation analysis

results are consistent with the findings of [Ahmat et al. \(2021\)](#), the findings validate that individuals who earned micro-credentials had higher levels of employability skills and competencies. Another study by [Desmarchelier and Cary \(2022\)](#) confer similar findings that micro-credentials can provide pathways to employment and career advancement by helping individuals develop specific skills and knowledge that are in demand in the job market. Thus, these programs are a substantial component in enhancing students' mobility and cultural capital.

The third hypothesis (**MCL → SOC**) affirms a positive relation between micro-credential learning and social capital ($\beta = 576, p < 0.000$); thus, **H5** is accepted. The outcomes of our study are corroborated by the observations of [De Janasz and Forret \(2008\)](#) that reveal micro-credential learning has a significant impact on social capital. The results corroborate with the findings of [Donald et al. \(2017\)](#), thus, micro-credential programs provide relevant exposure to students regarding the labor market. Therefore, activities that provide students with the chance to work on actual projects and collaborate with others can result in long-lasting and reliable connections; which will ultimately help in developing students' social capital.

The fourth hypothesis (**MCL → SCHC**) reveals a positive relation between micro-credential learning and scholastic capital ($\beta = 0.402, p < 0.000$). A study by [Lee \(2014\)](#) has determined similar results. The results indicate that micro-credential learning has a significant impact on scholastic capital. These learning affect the academic performance of students and provide the opportunity to enhance and augment academic performance. Similarly, [Donald et al. \(2017\)](#) determine scholastic capital can be developed via work-integrated projects which substantially improve and refine students' skills. Therefore, micro-credential learning provides a platform for students to strengthen their knowledge and skills regarding higher education or professional degree. Moreover, it provides opportunities for students to develop knowledge through work-integrated activities.

The fifth hypothesis (**CC → PE**) indicates a strong correlation between cultural capital and perceived employability ($\beta = 0.685, p < 0.043$). The results are consistent with the investigation of [Jäger \(2011\)](#), the finding concluded that cultural capital has a positive and significant impact on perceived employability. Moreover, the results are aligned with the study of [Caballero et al. \(2022\)](#), the findings established that individuals with higher levels of cultural capital, which includes knowledge, skills and other cultural activities, had higher perceived employability. The study also ascertained that cultural capital was a significant predictor of perceived employability, as employers demand a proactive attitude in employees ([Caballero et al., 2022](#)). Accordingly, cultural capital provides employees with all the mandatory cultural resources and the ability to exhibit them in their employment.

The sixth hypothesis (**SOC → PE**) demonstrates a positive relationship between social capital and perceived employability ($\beta = 0.348, p < 0.000$). The outcomes of our study are consistent with the study of [Zamudio et al. \(2014\)](#). The findings ascertain social capital significantly impacts perceived employability. Similarly, the findings determine social capital as the strongest predictor of graduates' perceived employability ([Agnihotri et al., 2022](#)). Moreover, the current study validates the network theory of social capital and [Bourdieu and Richardson \(1986\)](#), as it suggests that students can significantly avail advantages from social networks and connections. Thus, micro-credential learning helps students to interact with people and mentors that can guide them in their careers.

The seventh hypothesis (**SCHC → PE**) reveals a positive relation between scholastic capital and perceived employability ($\beta = 0.184, p < 0.000$). The results of [Tomlinson et al. \(2021\)](#) determine that scholastic capital has a positive connection with perceived employability. Correspondingly, another study by [Ho et al. \(2022\)](#) reveals the same findings. Hence, this helps employers to frequently examine knowledge attained through academic performance for assessing potential graduates; as social capital also helps applicants to acquire work experience. Therefore, this capital is especially useful for the students' academic and career achievement.

4.4.1 *Mediation analysis.* Table 7 demonstrates the results of mediating variables, i.e. micro-credential learning, social capital, scholastic capital, cultural capital and perceived employability. These mediating variables are referred to as hypotheses H8, H9, and H10, respectively.

The eighth hypotheses (**MCL → CC → PE**) reveal partial mediation of cultural capital between the relationship of micro-credential learning and perceived employability ($\beta = 0.084$, $p < 0.073$). The results of our research are supported by the findings of DiMaggio (1982) that validate cultural capital partially mediates the association between micro-credential learning and perceived employability. The findings highlight that cultural capital signifies students' inclination to participate in extracurricular activities to showcase their skills to potential employers (Ho *et al.*, 2022). As a result, when students engage in these activities driven by motivation, it enhances their self-confidence, self-awareness and self-consciousness. Thus, micro-credential learning also motivates students which increase students' confidence in their future careers and encourage investment in cultural capital.

The ninth hypotheses (**MCL → SOC → PE**) reveal partial mediation of social capital between the relationship of micro-credential learning and perceived employability ($\beta = 0.200$, $p < 0.000$). The results are similar to the result of Drenzo *et al.* (2015), the finding indicates that social capital partially mediates the association between micro-credential learning and perceived employability. Micro-credential learning influences perceived employability by giving students different means to increase their social capital by interacting with different people. While Caballero *et al.* (2021) and Tomlinson and Anderson (2021) also observed no mediation between career development and perceived employability of students. Therefore, it is evident that although micro-credentials may provide students with networks, they may struggle to effectively utilize and convert this capital into employment-related advantages. This challenge not only impacts students but also numerous other individuals seeking employment (Trimble and Kmec, 2011). Thus, social capital depicts partial mediation between the relationship of micro-credential learning and students' perceived employability.

The tenth hypothesis (**MCL → SCHC → PE**) reveals partial mediation of scholastic capital between the relationship of micro-credential learning and the perceived employability is accepted ($\beta = 0.074$, $p < 0.002$). Our results are consistent with the results of previous studies (Smith *et al.*, 2009), the findings established that scholastic capital partially mediates the association between micro-credential learning and perceived employability. Students can develop new skills and knowledge through micro-credential learning that can significantly boost their scholastic capital (Donald *et al.*, 2019). In addition, scholastic capital enables students to showcase their competencies and abilities to potential employers, ultimately improving their perceived employability. Moreover, micro-credential learning further strengthens students' scholastic capital via integrated learning which ultimately increases their perceived employability. Accordingly, the development of scholastic capital is a critical factor that connects micro-credential learning to perceived employability.

The current job market has witnessed a shift in employers' expectations of new hires, who are required to possess specific skills to enhance productivity. Simultaneously, students aspire to acquire skills rapidly to rejoin the workforce. In this regard, micro-credentialing presents a promising means to articulate the competencies that postsecondary institutions can offer to their graduating students (Gauthier, 2020). These programs can significantly contribute to students' training, familiarity with the future workplace, development of required skills and knowledge. Micro-credential learning supports students in career guidance and mentorship programs to guide their professional prospects. Thus, micro-credential programs play a significant role in advising and guiding students.

5. Conclusion

Globally, employers' expectations from new hires have changed in the current job market. They require specific skills that can increase productivity. Accordingly, students want to acquire these mandatory skills quickly to work efficiently and productively. In the current study, we examine the influence of micro-credential learning on students' perceived employability. This study aims to explore variables that help students to gain knowledge, enhance their careers and develop their human capital (social capital, cultural capital and scholastic capital). This research employs structural equation modeling and collects data through a questionnaire. The research examines the impact of micro-credential learning on students' perceived employability and their human capital, including social capital, cultural capital and scholastic capital. Additionally, the study investigates the mediating role of human capital (social capital, cultural capital and scholastic capital) between micro-credential learning and students' perceived employability.

The study ascertained the positive relationship of micro-credential programs with students' perceived employability. Moreover, the findings that micro-credential learning has a significant impact on students' human capital. Micro-credential learning positively influences students' cultural capital. Similarly, micro-credential learning positively influences students' social capital. Thus, students' scholastic capital is also significantly influenced by micro-credential learning. Furthermore, the findings indicate that human capital, i.e. cultural capital, social capital and scholastic capital partially mediated the relationship between micro-credential learning and students' perceived employability. Therefore, our findings stipulate that micro-credential learning is an essential component to improve students' perceived employability. Moreover, participation in micro-credential learning can ensure that students can identify diverse career directions, seek information about the labor market and educational system, attain relevant qualifications for their vocations and develop a plan for their future. Thus, the results demonstrate a significant relationship between micro-credential and perceived employability with social capital, scholastic capital and cultural capital.

5.1 Implications

Moreover, the results of the study have several useful implications. Firstly, education sector must create awareness among students regarding the importance of micro-credential learning. This awareness will increase students' confidence in their capacity/ability to get employment after graduation; the students get more mindful regarding the effects of micro-credential learning on their future. In addition, students will utilize their time in universities effectively to develop professional identities and human capital. Secondly, micro-credential learning augments human capital; hence, teachers should guide students to convert their capital into values beneficial for future employment by engaging them in effective and meaningful practices. Thirdly, the results will encourage students to actively participate in micro-credential learning activities and demonstrate a positive attitude toward learning advanced skills; for instance, investigating the labor market, communicating with employers about potential job requirements, and participating in work-integrated learning.

Moreover, the result proposed few recommendations for educational institutions, students and stakeholders to take essential steps toward students' career development. Educational institutions should introduce short courses, online learning courses and micro-credential courses that will help in developing students' skills and competencies. Additionally, stakeholders should design curricula that prioritize employability, utilize micro-credentialing methodology throughout academic program with involvement in the business sectors. Therefore, the findings of the study encourage students to improve their human capital, i.e. scholastic capital, cultural capital, and social capital by participating in different activities to

experience work-integrated learning. Moreover, a comprehensive and gradual approach that covers different subject areas can facilitate the development of student's abilities and sustained progress.

5.2 Recommendations and limitations

Few limitations of our study are stated as follows. Initially, the sample of study is restricted to one nation; thus, the sample limits how broadly the findings can be applied. Thus, future studies can use more varied samples and include cultural considerations in the research methodology. Secondly, the study solely examines students' perspective regarding employability. Moreover, future research can address these variables from several aspects such as teachers' perspective, integration of micro-credential programs in educational institutions and companies, acceptance of micro-credential programs, etc. Furthermore, the study has used a sample size of 324 students; hence, future studies can use large sample sizes to generalize and assess these characteristics extensively. Finally, data was collected through a questionnaire, specifically close-ended; thus, researchers can incorporate interviews, group discussions and open-ended surveys to attain comprehensive knowledge regarding their perceptions, feelings, attitudes, etc.

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